Path-Integral Approximation

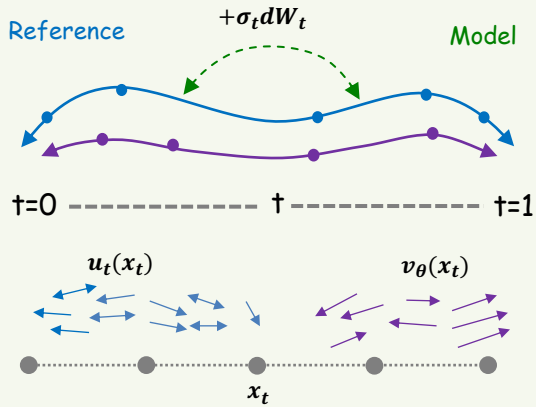
Exact $\log p_\theta(x_1|c)$
(Jacobian trace, high variance)

Variational path-space view

$$\log p_\theta(x_1|c) \geq \mathbb{E}_q[\log \frac{p_\theta(x_{0:1}|c)}{q(x_{0:1}|x_1)}]$$

Girsanov + variational path integral

$$\mathbb{E}_q[\log \frac{p_\theta(x_{0:1}|c)}{q(x_{0:1}|x_1)}] \approx -\int_0^1 \lambda_t || v_\theta(x_t) - u_t(x_t)||^2 dt$$



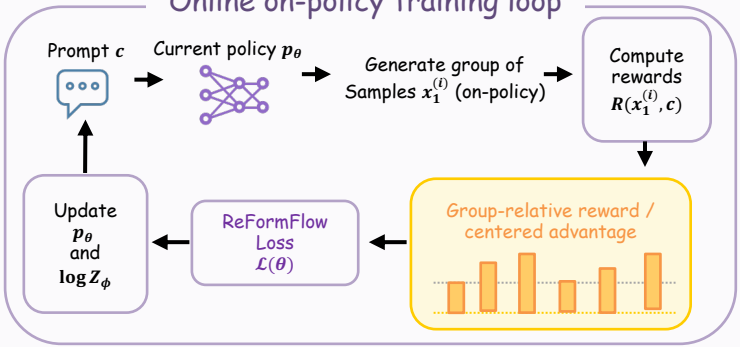
Turns density estimation into Tractable vector-field regression

Practical Optimization for ReFormFlow

Discrete objective used in practice

$$\mathcal{L}(\theta) = \mathbb{E}_{x_1,x_0}\left[\left(\log Z_\phi(c) + \frac{1}{K}\sum_{k=1}^{K} w_{t_k}\left(||v_{ref}(x_{t_k}) - u_{t_k}||^2 - ||v_\theta(x_{t_k}) - u_{t_k}||^2\right) - \frac{\tilde{R}(c,x_1)}{\gamma}\right)^2\right]$$

Online on-policy training loop



1.Adaptive scaling
Normalize reward via buffer std; stabilize KL

2.Learnable partition
MLP predicts Log from pooled Prompt embeddings

3.Group-relative rewards
Advantages distinguish better vs worse

4.CFG-free inference
Aligned policy directly generates high-quality samples